



Christina Ertural

✉ christina.ertural@quantumchemist.de

Nationality: German

in christina-ertural

Living in Berlin

R^G Christina-Ertural

QuantumChemist

PROFILE

Software Developer with scientific background

I am a (scientific) Software Developer and Full Stack Quantum Chemist with extensive experience in building and maintaining complex software workflows in C++ and Python. Currently, I am absolving a further training for data expert, advancing my skills in i.a. Python and SQL. My expertise spans open source development, (quantum chemical) algorithms, data processing, high-performance computing, and creating user-facing cli tools. I bring strong problem-solving skills from academia and the ability to adapt quickly to new technologies.

FURTHER TRAINING

06/2026 – 01/2027 Further training for Data Expert
StackFuel GmbH

- BI Specialist in SQL & Relational Databases:
Creation and administration of data bases, data-driven business decisions
- Data Analyst & Data Scientist:
Data analysis (EDA, SQL), predictions, statistics, machine learning and AI-supported decisions
- Project management:
Business Cases, planing resources, Kanban and Jira, management of risks, stakeholders and project quality

PROFESSIONAL EXPERIENCE

09/2022 – 08/2025 Senior Software Architect ML & Materials Informatics (Postdoctoral Researcher)
BAM Federal Institute for Materials Research and Testing, Berlin
(Governmental Research Institute)

- Designed and developed the Python workflow software AutoPLEX to automate large-scale data analysis and machine learning (ML) model generation for material simulations, within the Materials Project software frameworks and based on MongoDB database management, resulting in scientific publications, 150+ stars on GitHub, 20k+ downloads on pepy.tech and a large user base.
- Collaborated in interdisciplinary teams, translating scientific needs into robust code.
- Applied modern open source software engineering practices (version control Git/GitHub, testing, documentation); working on tools like atomate2, pymatgen, jobflow-remote and qtoolkit, which have a huge user base.

12/2017 – 07/2022 Software Developer / Doctoral thesis (Dr. rer. nat.)
RWTH Aachen University (Part of Germany's Excellence Initiative)

- C++ development and implementation of several quantum chemical bonding descriptors for the LOBSTER software (1.5k+ citations and huge user base) for electronic structure analysis, version control Apache Subversion SVN.
- Conducted computational research using VASP, Quantum ESPRESSO, ABINIT, resulting in publications with 2.5k+ citations.
- Supervised and trained students in quantum chemistry, coding and data analysis.

EDUCATION

12/2017 – 07/2022 Doctorate (Dr. rer. nat.), Theoretical Solid State Chemistry
RWTH Aachen University

10/2015 – 09/2017 Master of Science, Chemistry
RWTH Aachen University

10/2011 – 09/2015 Bachelor of Science, Chemistry
RWTH Aachen University

TECHNICAL SKILLS

Programming C++, Python, JavaScript/TypeScript, HTML/CSS, Git, GitHub, Linux, Bash scripting

Frameworks, Tools & More MongoDB, MySQL, HPC environments, Agile collaboration, Open Source Development,
Machine Learning, Workshop Organization, Mentoring

LANGUAGES

German (native), English (fluent), French (intermediate)